

# Revised Generation Interconnection System Impact Study Report

## For Queue Project Q22-2171

Kimballton 138 kV  
May 29, 2024

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2171. The Interconnection Customer proposes a 89.2 MW solar photovoltaic facility interconnecting at the Kimballton 138 kV substation in the West Virginia area. The request was received on August 30, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 541 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$255,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                           | Contingency               | Loading (%) | Rating (MVA) | DFAX (%) |
|----------------------------------------------|---------------------------|-------------|--------------|----------|
| Covadonga 1014-1013-1012-1011 / Peyton 138kV | Lowmoor / Hinton Ap 345kV | 104.7       | 405          | 54.4     |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                                     | Required Upgrade           | Cost Estimate | Allocation to Project |
|----------------------------------------------|----------------------------|---------------|-----------------------|
| Covadonga 1014-1013-1012-1011 / Peyton 138kV | Reconductor line           | \$1,827,000   | 14% (\$255,000)       |
|                                              | Total allocated to project |               | \$255,000             |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.